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R. N. G. Patel Institute of Technology (RNGPIT)

Isroli-Afwa, Bardoli

A Project Report

on

FACEGRAM(Share Photo using AI Face Recognition)

Under Subject of

DESIGN ENGINEERING - II – A (3150001)

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(Information Technology - IT Branch)

Submitted by:

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FACEGRAM(Share Photo using AI Face Recognition)

ABSTRACT

FACEGRAM is an innovative web and mobile-based photo-sharing platform designed to transform the way users manage, search, and share their digital images by leveraging advanced Artificial Intelligence for face recognition. The system automatically detects and identifies faces in uploaded photos, tagging individuals intelligently and enabling users to organize their photo collections efficiently without manual intervention. With AI-powered face recognition, FACEGRAM offers advanced features such as face-based search, personalized photo grouping, and smart sharing suggestions, allowing users to quickly locate images and share them with relevant contacts. The platform is developed using React and React Native for a responsive and interactive frontend, while PHP powers the backend, providing a robust and scalable server-side architecture to handle user data, image storage, and secure processing. AI integration is implemented with frameworks like OpenCV and DeepFace, ensuring accurate and real-time facial recognition even in large image libraries. FACEGRAM places strong emphasis on privacy and security, incorporating end-to-end encryption, user consent-driven tagging, and face-based access controls, giving users full control over their personal images and the individuals tagged in them. Beyond simplifying photo management, the platform enhances convenience, automation, and personalization, making the process of organizing and sharing photos smarter and more intuitive. By combining PHP backend development with modern frontend frameworks and AI technologies, FACEGRAM demonstrates a seamless integration of web development and artificial intelligence, delivering a secure, user-centric, and intelligent social sharing experience. It exemplifies how technology can enhance both usability and privacy in digital media platforms, offering a next-generation solution for photo sharing that aligns with the needs of today's visually driven, privacy-conscious users.

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CHAPTER 1: INTRODUCTION

1.1 Introduction to Domain

- This project operates within the Social Media and Artificial Intelligence (AI) domain, specifically focusing on intelligent photo sharing and management. The platform, FACEGRAM, leverages AI-based face recognition to simplify the organization, searching, and sharing of digital images. With the exponential growth of online social interaction and digital media consumption, users increasingly require tools that can manage large volumes of photos efficiently. FACEGRAM addresses this need by automatically detecting and tagging faces in uploaded images, enabling face-based searches, personalized grouping, and smart sharing suggestions. Developed for both web and mobile environments using React, React Native, and PHP for the backend, the platform integrates AI frameworks such as OpenCV and DeepFace to provide seamless, real-time facial recognition and photo management capabilities.

1.2 Importance to domain

- Efficient and intelligent photo management is crucial in today's visually driven digital world. Users frequently capture hundreds of photos, making manual organization time-consuming and error-prone. FACEGRAM ensures that users can easily locate and share images, enhancing convenience, personalization, and engagement on social media. Privacy and security are also critical, as users demand control over who can access their photos and how tagging is applied. By providing end-to-end encryption, consent-driven tagging, and face-based access control, FACEGRAM addresses these needs while improving user experience. The platform also benefits professional photographers, content creators, and businesses that rely on organized image libraries for marketing, documentation, or analytics purposes, making intelligent photo sharing a significant advancement in the social media and AI landscape.

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1.3 Team Selection

- The project team was chosen for their complementary expertise in full-stack development, AI integration, UX/UI design, and security. The team consists of four members: the Product Lead, who defines project scope and functional requirements; the Front-End Developer, proficient in React and React Native, responsible for creating a responsive and user-friendly interface; the Back-End Developer, specializing in PHP for server-side processing, database management, and secure API integration; and the UX Designer & QA/DevOps, who ensures optimal usability, performance, and security across devices. Together, the team brings the technical skills and domain understanding necessary to deliver a secure, intelligent, and user-centric photo-sharing platform.

Team ID: 907000

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Table 1.1 Team Members name and enrollment numbers

1.4 Observation

- During research on existing photo-sharing platforms, several limitations were observed. Most platforms require manual tagging of faces, making organization cumbersome. Many lack intelligent grouping or personalized sharing suggestions, limiting user convenience. Mobile usability is often suboptimal, reducing accessibility for users on smartphones or tablets. Privacy and security issues are prevalent, with some apps storing images or facial data insecurely or offering limited control over tagging and sharing. Additionally, platforms rarely provide automated notifications, such as alerts when a specific person appears in new photos. FACEGRAM addresses these challenges by providing AI-driven face recognition, real-time intelligent tagging, secure data handling, responsive design, and customizable sharing options, offering a smarter, safer, and more efficient photo-sharing experience.

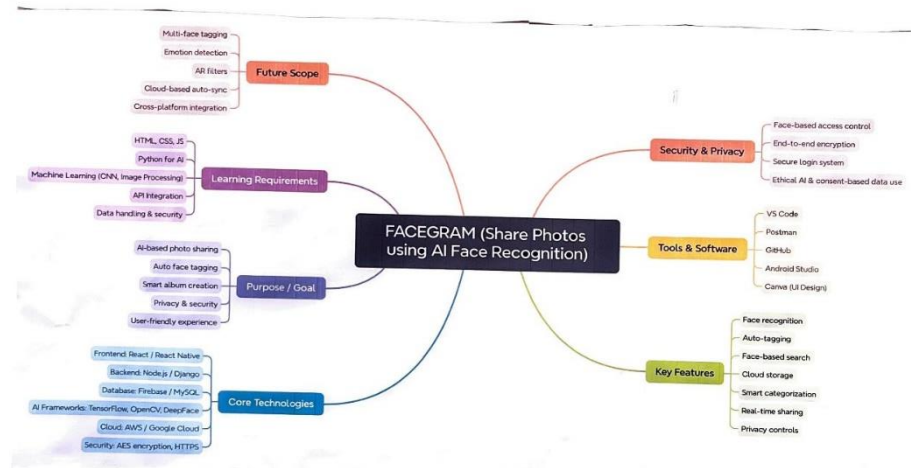
Chapter 2 Canvas and Framework

2.1 Mind Mapping (Data Analysis)

A Mind Map is a visual tool used to organize and represent the core ideas and components of the FACEGRAM project. It begins with the central concept—FACEGRAM, an AI-based photo-sharing platform—placed at the center, from which all major elements of the project branch out. These primary branches represent the key aspects of the system such as Artificial Intelligence Integration, Frontend Development, Backend Development, Privacy and Security, User Experience (UX/UI), and Core Features. Each of these major branches further divides into sub-branches illustrating specific details, like AI technologies (OpenCV, DeepFace), backend logic (PHP, database, APIs), and user-focused functionalities (face recognition, tagging, search, sharing, and notifications). The Mind Map provides a hierarchical and interconnected overview of FACEGRAM, helping visualize how each component contributes to the platform's goal of creating a smart, secure, and user-friendly photo-sharing experience. It serves as a visual thinking tool that enables the project team to structure ideas clearly, understand relationships among modules, and identify dependencies between technical and functional aspects. By analyzing and organizing these ideas in a single diagram, the team can better comprehend the system's architecture, streamline development, and ensure effective collaboration. In essence, the FACEGRAM Mind Map helps synthesize complex information into a simple and meaningful visual format, making it easier to plan, recall, and innovate throughout the project's lifecycle. It not only aids in conceptual clarity but also enhances creativity and problem-solving by visually connecting AI technology, design, and user interaction into one cohesive framework..

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Mind Mapping



[Fig-2.1.1-Mind Map]

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2.2 AEIOU Summary Framework

AEIOU is an investigative tool to help interpret observations gathered by ethnographic practices in the field. It is an Observation tool. Its two primary functions are to code data, and to develop building blocks of models that will ultimately address the objectives and issues of a client.

➤ **A-Activities:**

Activities are goal directed sets of actions – paths towards people want to accomplish.

- Upload
- Share
- Detect
- Tag
- Auto album
- Search
- Manage privacy

➤ **E-Environment:**

Environments include the entire arena where activities take place. In this section, we have added the environment in which the situations are taking place and the atmosphere which is creating a problem inside or outside of your domain.

- Web app
- Mobile app
- Cloud storage
- Real-time AI
- User-friendly UI
- Home / Events / Office

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➤ **I-Interactions:**

Interactions are between person and someone or something else. In this section, we have added the interaction happening in the current system.

- User ↔ AI
- Upload & view
- Auto tagging
- Notifications
- Privacy control

➤ **O- Objects:**

What are the objects and devices people have in their environments and how do they relate to their activities.

- Photos
- Profiles
- AI model
- Database
- Dashboard
- Devices

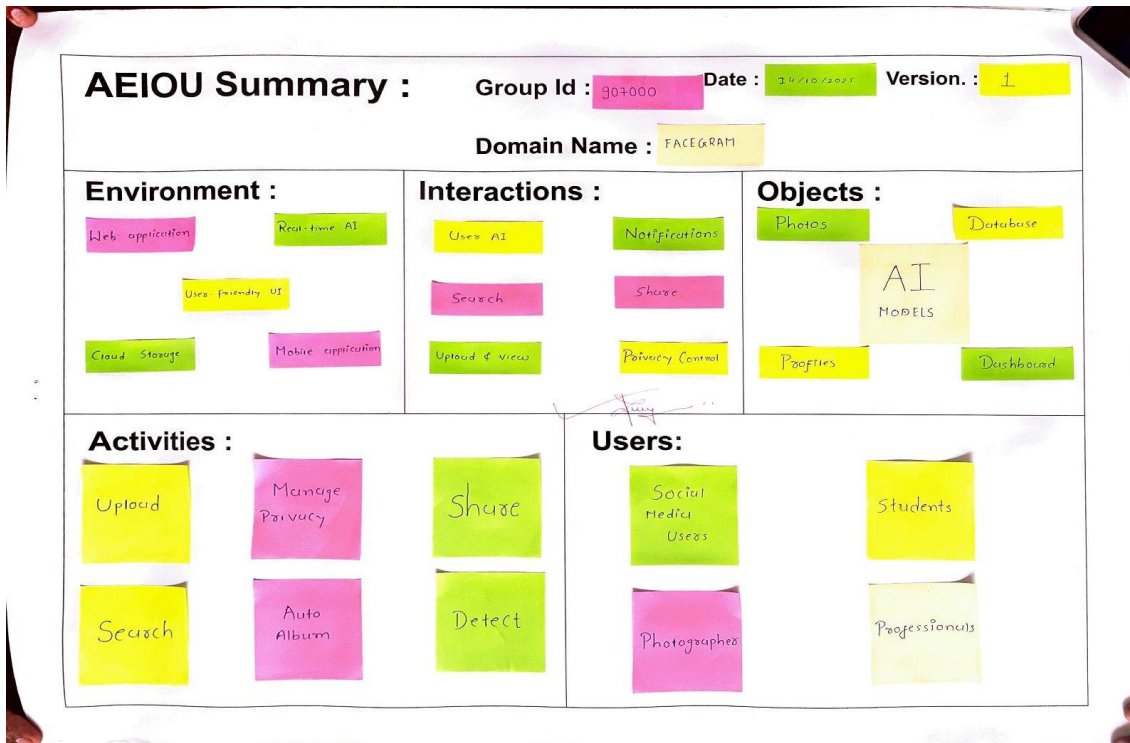
➤ **U –Users:**

Users are the people whose behaviours, preferences, and needs are being observed.

- Social media users
- Families & friends
- Photographers
- Students
- Professionals
- Privacy-focused users

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➤ AEIOU Summary canvas:-



[Fig-2.2.6.1-AEIOU Summery Canvas]

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2.3 Empathy mapping canvas

Meaning of empathy is understanding and entering to another's feelings. The empathy mapping is a great powerful tool that helps putting yourself inside the head of a person you might be looking at a product user. It allows you to quickly grasp their experience and where they are coming from.

➤ **Users:**

- Social media users
- Families & friends
- Photographers
- Students & professionals

➤ **Stakeholders:**

- End users
- Developers & AI engineers
- Cloud service providers
- Security & privacy experts
- Project investors & admins

➤ **Activities:**

- Upload & share photos
- AI detects & tags faces
- Auto smart album creation
- Face-based photo search
- Manage privacy settings
- Share albums securely
- Receive tag notifications

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➤ Story Boarding:

Happy Story:

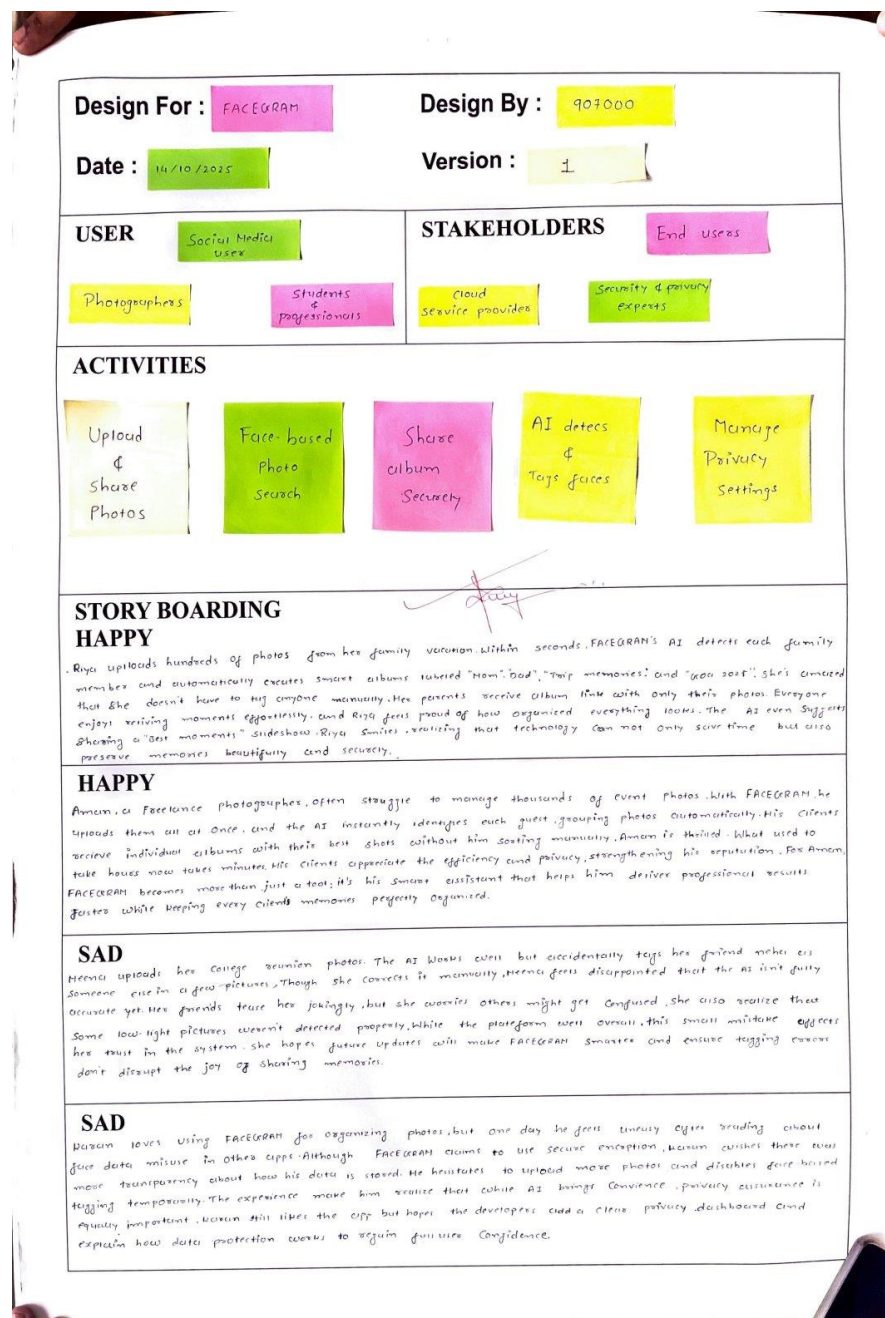
1. Riya uploads hundreds of photos from her family vacation. Within seconds, FACEGRAM's AI detects each family member and automatically creates smart albums labeled "Mom," "Dad," "Trip Memories," and "Goa 2025." She's amazed that she doesn't have to tag anyone manually. Her parents receive a private album link with only their photos. Everyone enjoys reliving moments effortlessly, and Riya feels proud of how organized everything looks. The AI even suggests sharing a "Best Moments" slideshow. Riya smiles, realizing that technology can not only save time but also preserve memories beautifully and securely.
2. Aman, a freelance photographer, often struggles to manage thousands of event photos. With FACEGRAM, he uploads them all at once, and the AI instantly identifies each guest, grouping photos automatically. His clients receive individual albums with their best shots without him sorting manually. Aman is thrilled — what used to take hours now takes minutes. His clients appreciate the efficiency and privacy, strengthening his reputation. For Aman, FACEGRAM becomes more than just a tool; it's his smart assistant that helps him deliver professional results faster while keeping every client's memories perfectly organized.

Sad Story:

1. Meena uploads her college reunion photos. The AI works well but accidentally tags her friend Neha as someone else in a few pictures. Though she corrects it manually, Meena feels disappointed that the AI isn't fully accurate yet. Her friends tease her jokingly, but she worries others might get confused. She also realizes that some low-light pictures weren't detected properly. While the platform performs well overall, this small mistake affects her trust in the system. She hopes future updates will make FACEGRAM smarter and ensure tagging errors don't disrupt the joy of sharing memories.

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- Karan loves using FACEGRAM for organizing photos, but one day he feels uneasy after reading about face data misuse in other apps. Although FACEGRAM claims to use secure encryption, Karan wishes there was more transparency about how his data is stored. He hesitates to upload more photos and disables face-based tagging temporarily. The experience makes him realize that while AI brings convenience, privacy assurance is equally important. Karan still likes the app but hopes the developers add a clear privacy dashboard and explain how data protection works to regain full user confidence.



[Fig-2.3.4.1-Empathy Canvas]

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2.4 Ideation Canvas

An ideation canvas is a rough whiteboard where ideas can be stretched into any limits or dimensions. Ideation phase is not aimed at finding solutions to the define problem but to define the best possible problem and stretch out its possible scope. The field is set and the overall agenda is to build the clones of the ideas and pivot them throughout the canvas to discover new possibilities.

➤ **People:**

- Social media users
- Families & friends
- Photographers
- Students
- Tech enthusiasts

➤ **Activities:**

- Upload photos
- AI face detection
- Auto-tagging
- Smart albums
- Search by face
- Privacy control
- Sharing

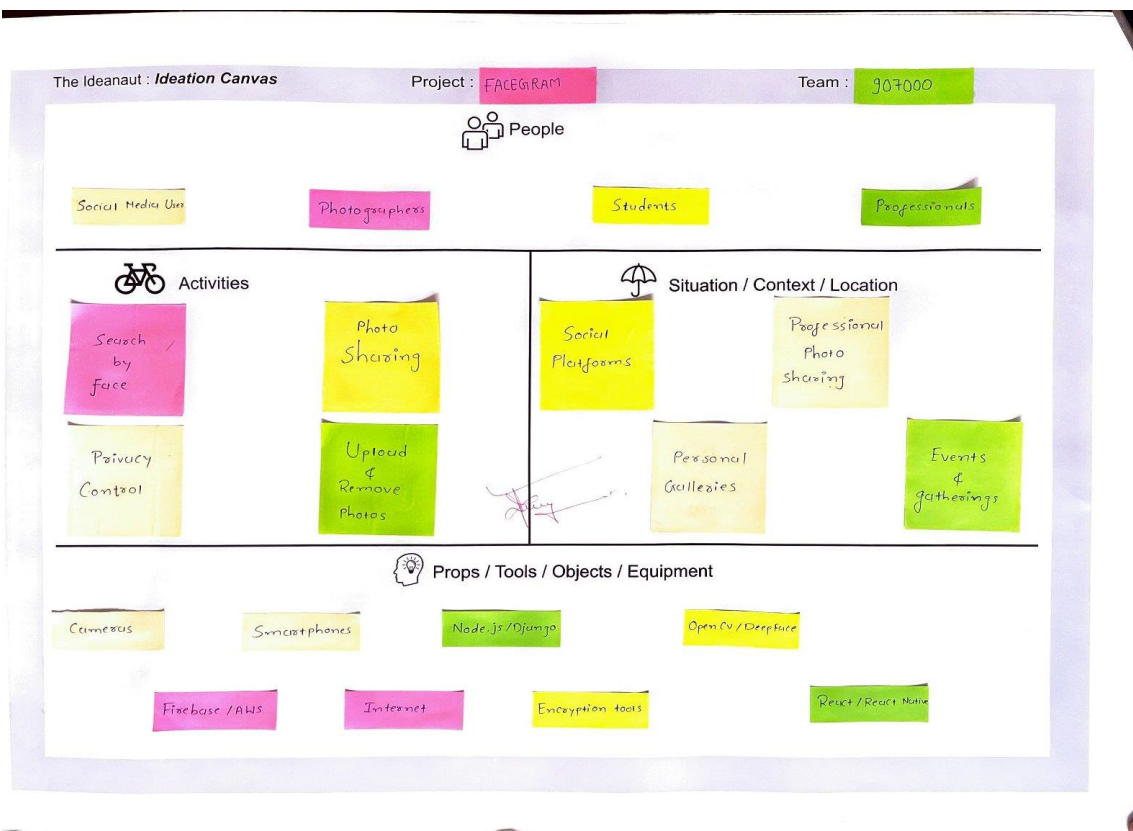
➤ **Situation/Context/Location:**

- Web & mobile use
- Social platforms
- Events & gatherings
- Personal galleries
- Professional photo sharing

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➤ Props/Possible Solution:

- Smartphones
- Cameras
- Internet
- React / React Native
- Node.js / Django
- Firebase / AWS
- OpenCV / DeepFace
- Encryption tools



[Fig-2.4.4.1-Ideation Canvas]

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2.5 Product Development Canvas

The product development of canvas is totally based on specific product. That is, how product is useful to users which facilities and functions provided by that product and how it differs from recent market products. We should make attempt what is the new idea behind the product and in this section, we can introduce. The new idea and its evolution will reduce the human effort and help them to go easy with life and work.

➤ **Purpose:**

- AI photo sharing
- Auto face recognition
- Smart organization
- Privacy & security
- User convenience

➤ **People:**

- Users: Students, families, photographers
- Stakeholders: Developers, testers, AI engineers
- Beneficiaries: Social media users, digital photo sharers

➤ **Components:**

- Frontend: React / React Native
- Backend: Php
- Database: MySQL
- AI: TensorFlow / OpenCV
- Cloud: AWS / Google Cloud
- Security: AES, HTTPS

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➤ **Product Functions:**

- Upload photos
- Detect faces (AI)
- Auto-tagging
- Face search
- Secure sharing
- Privacy settings

➤ **Product Features:**

- DeepFace/OpenCV AI
- Smart categorization
- Face-based access
- Encrypted cloud storage
- React UI / responsive
- Real-time alerts

➤ **Product Experience:**

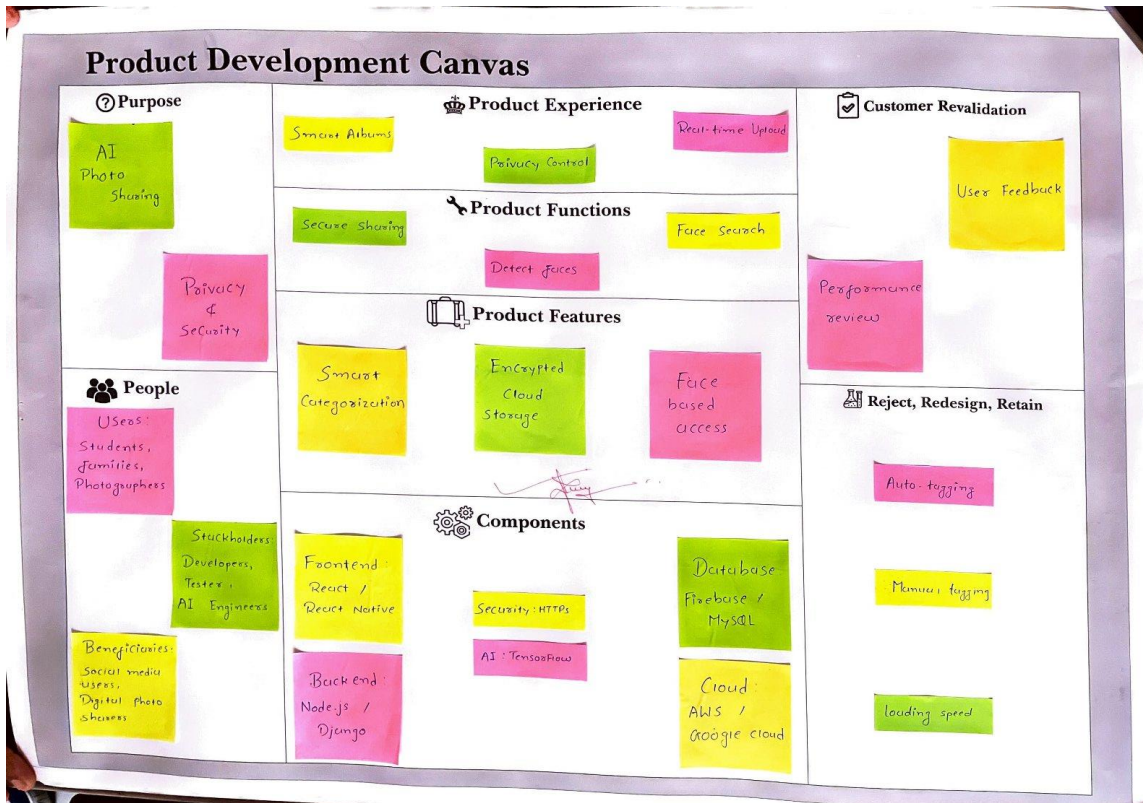
- Auto-tag photos
- Smart albums
- Face-based login
- Easy navigation
- Privacy control
- Real-time upload

➤ **Customer Revalidation**

- User feedback
- Ease of use
- Accuracy testing
- UI/UX improvements
- Performance review

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[Fig-2.5.8.1-Product Development Canvas]



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2.6 Learnings Need Matrix (LNM)

A Learning Needs Matrix is a structured tool that helps identify, organize, and prioritize learning requirements within an organization or for individuals. It compares current skills with required competencies, highlighting gaps that need development. By using this matrix, organizations can efficiently allocate resources, plan targeted training programs, and support continuous growth, ultimately enhancing overall performance and productivity.

➤ Tools & Skills:

- Python, OpenCV, TensorFlow
- React, Node.js
- Firebase, AWS
- VS Code, Postman
- GitHub, JSON

➤ Standards & Principles:

- IEEE Standards
- ISO 27001
- GDPR Privacy
- UI/UX Guidelines
- Ethical AI

➤ Materials / Components:

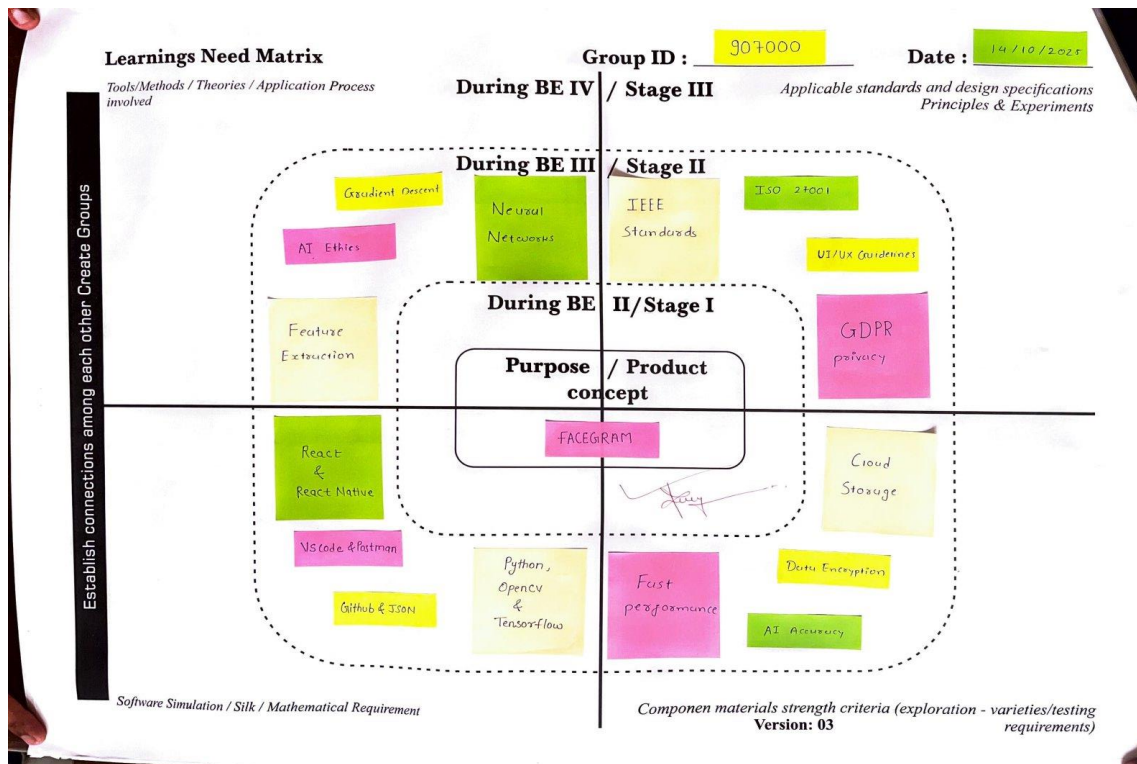
- Cloud Storage
- AI Accuracy
- Data Encryption
- Cross-Platform
- Fast Performance

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➤ Math & Theory:

- IEEE Standards
- ISO 27001
- GDPR Privacy
- UI/UX Guidelines
- Ethical AI

[Fig-2.6.5.1-Learnings Need Matrix]

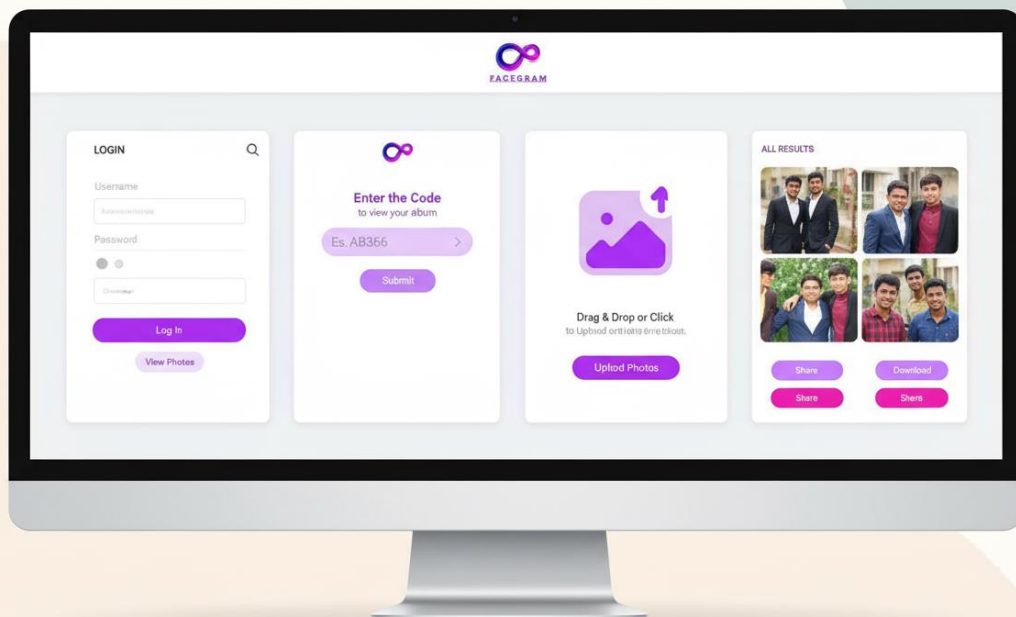


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Chapter 3 Rough Prototype

3.1 Rough Prototype Design

Demo



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CONCLUSION

FACEGRAM represents a significant advancement in the way users interact with and manage their digital photos by combining the power of Artificial Intelligence with modern web and mobile technologies. Through intelligent face recognition, automated tagging, and smart sharing features, the platform simplifies the photo organization process and enhances user convenience. By integrating frameworks such as OpenCV and DeepFace, FACEGRAM delivers accurate and efficient face detection, enabling users to easily search, group, and share photos based on identified individuals. The use of PHP for backend development ensures reliable server-side processing and secure data handling, while React and React Native provide a seamless, responsive, and user-friendly interface across devices. Beyond its technical capabilities, FACEGRAM also emphasizes privacy, data protection, and user control, implementing end-to-end encryption and consent-based tagging to ensure that users' personal data remains secure. The platform demonstrates how AI can be effectively harnessed to improve everyday digital experiences, merging automation with user-centric design. Overall, FACEGRAM not only simplifies digital photo management but also redefines social photo sharing by making it smarter, safer, and more personalized. This project showcases the potential of integrating AI into social media applications, laying a strong foundation for future enhancements such as emotion-based categorization, real-time photo recommendations, and cloud-based collaborative sharing.

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